Evan Gattoni

Car Crash Survivability Logistic Regression

Introduction

Motor vehicle crashes cause thousands of deaths every year in America, and understanding the factors associated with what makes a deadly crash helps keep more people safe. This project analyzes a crash test simulation dataset, using categorical analysis and a logistic regression model within the GLM framework to understand the factors that contribute to surviving a car accident. The primary objective of this analysis is to model the probability of survival following a high-speed accident and assess the correlations between survival and several predictors. The response variable is survival status; this number is a binary (0 = died, 1 = survived). The nature of this variable makes logistic regression the ideal model choice. Additionally, this analysis evaluates independence within categorical variables, compares nested models, and conducts model adequacy tests. This model and analyses were all completed in R.

Data Description

The dataset contains 200 observations, each corresponding to whether an individual test dummy survived in a crash test simulation. The following variables were used in the analysis:

- Survived (0 = died, 1 = survived)
- Age (continuous)
- Gender (categorical)
- Speed of impact (continuous)

- Helmet used (Yes/No; categorical)
- Seatbelt used (Yes/No; categorical)

Before analysis, some data cleaning steps were taken to ensure the model would run smoothly. Categorical variables were explicitly converted to factor variables, and reference levels were also assigned. Male was the baseline for gender, and “No” for seatbelt and helmet usage. These baselines allow for the other coefficients to be compared against the baseline groups for easy analysis. A few observations were removed due to the lack of some categorical values; this is not uncommon in real-world datasets, so it is not any trouble in our analysis.

Exploratory Categorical Analysis

Independence between survival and the categorical predictors was checked using chi-square tests. Across all the categorical variables in the dataset, no statistically significant association was noted. Helmet usage ($\chi^2 = 0.196$, $p = 0.658$), seatbelt usage ($\chi^2 = 0.484$, $p = 0.487$), and gender ($\chi^2 = 3.79$, $p = 0.150$) all had p-values greater than .05, so we can assume that these categorical variables do not hold a strong association with survival just based on these tests.

Model Specification and Selection

Considering we have a binary response variable, logistic regression makes sense as the primary model. Logistic regression is a GLM that relates predictors to the log-odds of the outcome, in this case, survival. A full logistic regression model was constructed, allowing evaluation of demographic factors and crash-related statistics. To address the issue of selecting optimal independent variables, a reduced logistic regression model was also made available, focusing on only speed, helmet usage, and seatbelt usage. This model is to assess the effectiveness of the safety measures. Comparing the full model to this reduced model will give us insight into how the demographic variables contribute to survival. Model comparison was done with two tests, a likelihood ratio test and an AIC test. Likelihood ratio judges whether an increase in model complexity improves the fit of the model, while AIC balances model fit and parsimony.

Results

Several predictors showed a marginal association with survival, while some showed no association:

- Age showed a positive association with survival ($p = .069$). A coefficient of 1.02 would suggest that each additional year of age relates to a 2% increase in survival odds. This would not make sense because we would expect extremely elderly people to have lower survival odds. However, this effect is present in the dataset to a relatively significant extent.
- Gender showed an odds ratio of .59 ($p = .071$), which indicates females have lower odds of survival than males. This p-value tells us there is a marginal association between gender and survival.

- Seatbelt usage and helmet usage showed similar results to each other. Both safety factors had positive odds ratios, which would suggest that using safety features makes survival more likely. However, due to high p-values, we cannot consider this correlation as statistically significant.
- Surprisingly, the speed of impact showed no meaningful association with survival.

No single predictor seemed to dominate the model; a few variables had some moderate association with survival, while some did not at all. Because of the real-world implications of a car crash, there are so many more factors that play into it besides just the ones in the dataset, so it is to be expected that we do not see extreme correlations.

After looking at the likelihood ratio test comparing the two models, we see a result that indicates that adding age and gender to the model significantly improves the model fit (χ^2 = 8.02, $p = 0.046$). The AIC also favored the full model over the reduced model. Considering that age and gender were the two most significantly associated predictors, it makes sense that they improve the overall model. Multicollinearity was checked using VIF, and since all VIF values were close to 1, there is no evidence showing an issue with collinearity.

Conclusion

In this analysis, a logistic regression model was used to analyze survival outcomes in crash dummy tests. While no predictor emerged as a primary significant predictor, the analysis identified some moderate associations between some factors and survival. The results would point towards several interacting factors in combination that would explain survival, rather than our oversimplified data. The full logistic regression had a more accurate fit compared to a reduced specification model, stating the value of demographic variables in a dataset like this.

Overall, this analysis is a good representation of the strength of logistic regression when having a binary predictor, when it is partnered with smart data processing and model comparison.

References

https://www.kaggle.com/datasets/himelsarder/road-accident-survival-dataset